

Drill Problems 10

*Author: Jaehoon Song**Release: 2025-08-10 15:09:34-04:00***Purpose and Usage:**

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C O L U M B I A A C A D E M Y

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Written by Jaehoon Song

1. **Solving for an Expression** (10 points)

Given that $2(x + 4) = 18$, what is the value of $x + 6$?

- A. -4
- B. 5
- C. 9
- D. 11

Answer:



2. **Inequality Solution** (10 points)

$$3x + 2 < -5(x + 6)$$

Which of the following values is a solution to the given inequality?

- A. -5
- B. -4
- C. 3
- D. 4

Answer:



3. Mean from Bar Graph (10 points)

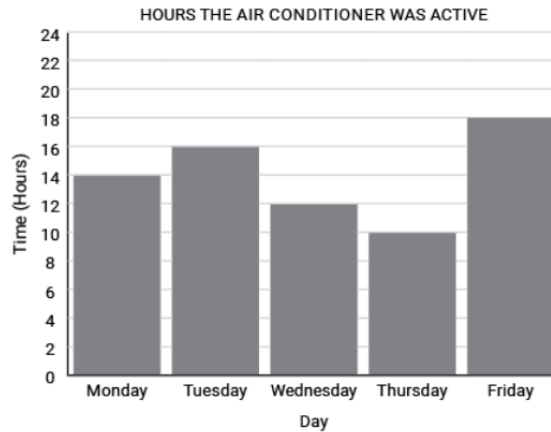


Figure 1: reference attached

The bar graph shows how many hours the air conditioner was on from Monday to Friday in a given week. What is the mean number of hours the air conditioner was on during this period?

- A. 8
- B. 10
- C. 12
- D. 14

Answer:



4. Quadratic Expansion Constant (10 points)

The expression $-2(x + 3)^2 + 6$ is equivalent to $ax^2 + bx + c$, where $a < 0$ and $b < 0$. What is the value of c ?

Answer:



5. Slope of Perpendicular Line (10 points)

What is the value of the slope of a line perpendicular to the line $5x - 6y + 30 = 0$?

Answer:



6. **Sum of Quadratic Roots** (10 points)

$$3x^2 + 9x - 27 = 0$$

If m and n are solutions to the given equation, what is the value of $m + n$?

Answer:



7. **Point on a Circle** (10 points)

The graph of $(x - 3)^2 + y^2 + 8y = 84$ is a circle in the xy -plane. Which of the following coordinates lie on the circle?

- A. $(1, 7)$
- B. $(-2, 5)$
- C. $(-3, 4)$
- D. $(3, -6)$

Answer:



8. **Stockpile Percentage Remaining** (10 points)

At the start of the winter, the US stockpile of corn was 5 billion bushels. Two months later, $p\%$ of the stockpile had been used. Which expression best represents the number, in billions, of bushels of corn remaining in the US stockpile at the end of those two months?

- A. $2(5) \left(\frac{100-p}{100} \right)$
- B. $5 \left(\frac{100-p}{100} \right)$
- C. $\left(\frac{p}{100} \right)$
- D. $2(5)(100 - p)$

Answer:



9. **Profit Function Interpretation** (10 points)

The equation $P(y) = 37(1.1)^y$ gives the estimated annual profit, in thousands of dollars, at a pottery studio, where y is the number of years since the studio moved to a new location. Which of the following is the best interpretation of the number 37 in this context?

- A. The estimated annual profit, in thousands of dollars, when the studio moved to the new location
- B. The increase in the estimated annual profit, in thousands of dollars, each year
- C. The number of years since the studio moved to the new location
- D. The percent increase in the estimated annual profit each year

Answer:



10. **Isosceles Triangle Angle** (10 points)

Triangle JKL is an isosceles triangle. The measure of angle J is 30° . If JL is the longest side of the triangle, what is the measure of angle K ?

- A. 30
- B. 75
- C. 120
- D. 150

Answer:



11. **Function Value from Rule** (10 points)

If $f(2x) = 9x - 7$, what is the value of $f(6)$?

- A. 11
- B. 20
- C. 47
- D. 101

Answer:



12. **System of Equations Solutions** (10 points)

$$3x - 4y = 16$$

$$-6x = -8y + 32$$

How many solutions does the given system of equations have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Answer:



13. **Solving for Product** (10 points)

If $\frac{5}{x} = \frac{2}{y}$ and $\frac{4}{x} - \frac{2}{y} = -\frac{1}{5}$, what is the value of xy ?

Answer:



14. **Radical and Exponent Equation** (10 points)

In the equation $\frac{1}{2}\sqrt{a} = 2\sqrt[3]{b}$, both a and b are positive real numbers. If $a = 8$ and the equation is written in the form $a^x = b$, what is the value of x ?

Answer:

□

15. **Exponential Function from Points** (10 points)

The function g is defined by $g(x) = a^x - b$, where a and b are constants. In the xy -plane, the graph of $g(x)$ passes through the points $(0, -6)$ and $(1, 2)$. What is the value of a ?

- A. 6
- B. 7
- C. 8
- D. 9

Answer:

□

16. **Quadratic Solution with Radical** (10 points)

If $1 + \frac{a\sqrt{2}}{2}$ is a solution to the equation $2x^2 - 4x - 7 = 0$ and $a > 0$, what is the value of a ?

Answer:

□

17. **Transformation of Quadratic Function** (10 points)

If $f(x) = 2(x - 3)^2 + 8$ is transformed into $g(x) = 2(x - 5)^2 + 5$, which of the following describes the transformation?

- A. The x -coordinate moves to the right 2 units and the y coordinate moves 3 units down.
- B. The x -coordinate moves to the left 2 units and the y -coordinate moves 3 units down.
- C. The x -coordinate moves to the right 2 units and the y coordinate moves 3 units up.
- D. The x -coordinate moves to the left 2 units and the y -coordinate moves 3 units up.

Answer:

□

18. **Unit Conversion: Density** (10 points)

A sample of dried yellow pine has a density of 420 kilograms per cubic meter. To the nearest tenth, what is the density, in pounds per cubic foot, of this sample? (Use 1 kilogram = 2.2 pounds and 1 meter = 3.3 feet)

- A. 5.3
- B. 25.7
- C. 127.3
- D. 280.0

Answer:



19. **Arc Length in Circle** (10 points)

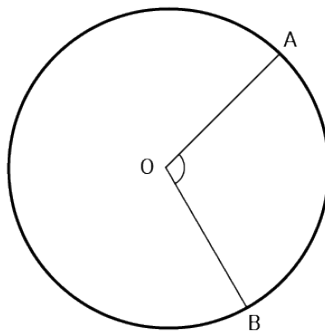


Figure 2: reference attached

In the given circle, the radius is 6 centimeters and angle AOB measures $\frac{7}{12}\pi$ radians. What is the value of the arc length AB , in centimeters?

- A. $\frac{7}{2}\pi$
- B. $\frac{21}{2}\pi$
- C. 12π
- D. 36π

Answer:



20. **Right Triangle Altitude Length** (10 points)

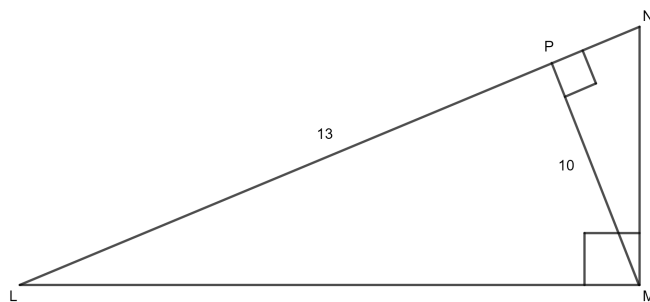


Figure 3: reference attached

In the image shown, right triangle LMN has a right angle at M . Altitude PM is shown and $\angle NPM$ is a right angle. $LP = 13$ and $PM = 10$. Which of the following are closest to the length of NP ?

- A. 3.0
- B. 6.2
- C. 7.1
- D. 7.7

Answer:



21. **Quadratic from Table: Find b** (10 points)

$$f(x) = 24x^2 + bx + c$$

For the given function f , b and c are constants, and the graph of $y = f(x)$ in the xy -plane passes through the points in the table below. What is the value of b ?

x	$f(x)$
$-\frac{15}{8}$	0
0	c
$\frac{11}{4}$	0

Answer:



22. **Polynomial Factors from Table** (10 points)

Select values for the polynomial function $g(x)$ are shown in the table. Based on the values in the table,

x	$g(x)$
-7	10
-4	0
0	5
3	1
5	-6
11	0

which of the following must be factors of $g(x)$?

- A. $(x - 5)$
- B. $(x + 5)$
- C. $(x - 4)$ and $(x + 11)$
- D. $(x + 4)$ and $(x - 11)$

Answer:



23. **Absolute Value Expression** (10 points)

If $|7x + 14| = 98$ and x is positive, what is the value of $x + 2$?

Answer:

□

24. **Linear Function from Table** (10 points)

The table above shows some values of the linear function f . Which of the following defines f ?

Values of the Linear Function f

a	3	6	9	12
$f(a)$	-2	-1	0	1

- (A) $f(a) = \frac{1}{3}a - 3$
- (B) $f(a) = -\frac{1}{3}a - 1$
- (C) $f(a) = a - 5$
- (D) $f(a) = 2a - 8$

Answer:

□

25. **Plant Growth** (10 points)

Stan's plant grows 2.5 inches per week. It is 15 inches tall now. In how many weeks will the plant be 42.5 inches tall?

Answer:

□

26. **Number of Solutions to an Equation** (10 points)

$$3(x - 2) - 2(x - 1) = -x + 2(x + 2) - 8$$

How many solutions does the given equation have?

- (A) Zero
- (B) Exactly one
- (C) Exactly two
- (D) Infinitely many

Answer:

□

27. **Function Value from Shifted Input** (10 points)

For $f(x + 3) = \sqrt{x^3} + 17$, what is the value of $f(4)$?

- (A) 18
- (B) 25
- (C) 28
- (D) 35.5

Answer:



28. **Percent Increase Multiplier** (10 points)

Theresa invested in a company, and the value of her investment increased by 9% from 2021 to 2022. If the value of her investment in 2022 is k times the value it was in 2021, what is the value of k ?

- (A) 0.09
- (B) 0.9
- (C) 1.09
- (D) 1.9

Answer:



29. **Possible Values of Expressions** (10 points)

Which of the following expressions can equal -1 for some value of a ?

- (A) $|a - 3| + 2$
- (B) $|a - 2| + 1$
- (C) $|a - 1|$
- (D) $|a - 2| - 2$

Answer:



30. **System of Equations Solution** (10 points)

$$\frac{m}{n} = 36(n - 4) = m$$

The solution to the given system of equations is (m, n) . What is the value of n ?

- (A) 4
- (B) 8
- (C) 16
- (D) 24

Answer:



31. **Arc Length and Central Angle** (10 points)

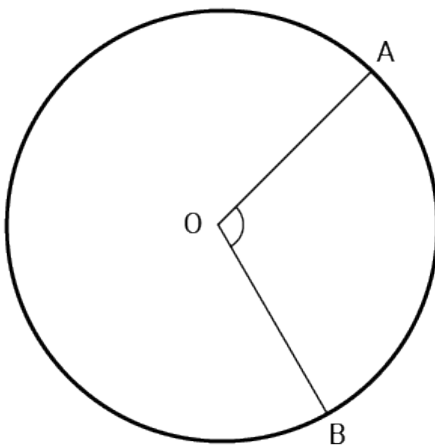


Figure 4: reference attached

In circle O , the length of AO is 12 units and the length of minor arc AB is 8π . What is the measure, in degrees, of $\angle AOB$?

Answer:



32. **Algebraic Expression Simplification** (10 points)

Which expression is equivalent to $\frac{x+4}{2x+2} + \frac{5}{x+1}$?

- (A) $\frac{x+7}{x+1}$
- (B) $\frac{x+9}{3x+3}$
- (C) $\frac{x+14}{2x+2}$
- (D) $\frac{5x+20}{2x^2+4x+2}$

Answer:



33. **Function Translation** (10 points)

Which of the following is the equation of $g(x) = 2x$ when it is moved 1 unit to the left and 1 unit up?

- (A) $y = 2x - 1$
- (B) $y = 2x + 1$
- (C) $y = 2x + 3$
- (D) $y = 2x + 5$

Answer:



34. **Parallel Lines and Angles** (10 points)

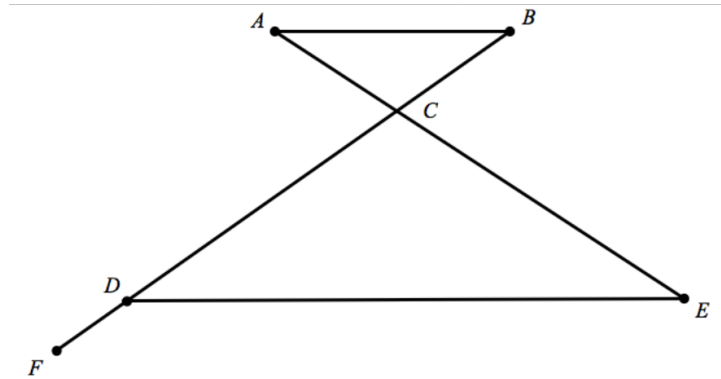


Figure 5: reference attached

If $\overline{AB} \parallel \overline{DE}$, $AC = BC$, and the measure of $\angle ACB = 88^\circ$, what is $m\angle FDE$?

- (A) 88°
- (B) 92°
- (C) 128°
- (D) 134°

Answer:



35. **Linear Functions Comparison** (10 points)

The function $f(x) = mx + b$ and the function $g(x) = nx + b$ are defined for all real values of x . Which statement is true about the functions?

- (A) If $|m| > |n|$, then $f(x) > g(x)$ for all values of x .
- (B) If $|m| = |n|$, then $f(x) = g(x)$ for all values of x .
- (C) If $m > 0$, then $f(x) > g(x)$ for all values of x .
- (D) If $m \neq n$, then $f(x) = g(x)$ for only one value of x .

Answer:



36. **Sum of Tangents in Right Triangles** (10 points)

Triangles ABC and BCD are right triangles, where $\angle A$ and $\angle D$ are right angles. If $AB = AC$ and $BD = CD$, what is the value of $\tan(\angle ABC) + \tan(\angle DBC)$?

Answer:



37. **Geometric Sequence Formula** (10 points)

The first term in a sequence is 21. Each term after the first term is 1.5 times the preceding term. If s represents the n th term of the sequence, which equation gives s in terms of n ?

- (A) $s = 1.5 (21^n)$
- (B) $s = 1.5 (21^{n-1})$
- (C) $s = 21 (1.5^n)$
- (D) $s = 21 (1.5^{n-1})$

Answer:



38. **Standard Deviation Change** (10 points)

The table above shows the distribution of test scores in Mrs. Keaton's English class. Which of the following

Vocabulary Test Scores

Test Score	Number of Students
100	1
90	5
80	10
70	8
60	2

sets of test scores, if added, would change the standard deviation of the distribution the least?

- (A) 60, 60, 60
- (B) 60, 100, 100
- (C) 70, 80, 90
- (D) 90, 90, 90

Answer:



39. **Translation of Parabola and x -Intercept** (10 points)

The graph of $2y - 2x^2 + 14x = -36$ is translated to the left 3 units in the xy -plane. What is the positive x -coordinate of the x -intercept of the resulting graph?

Answer:



40. **Polynomial from Graph** (10 points)

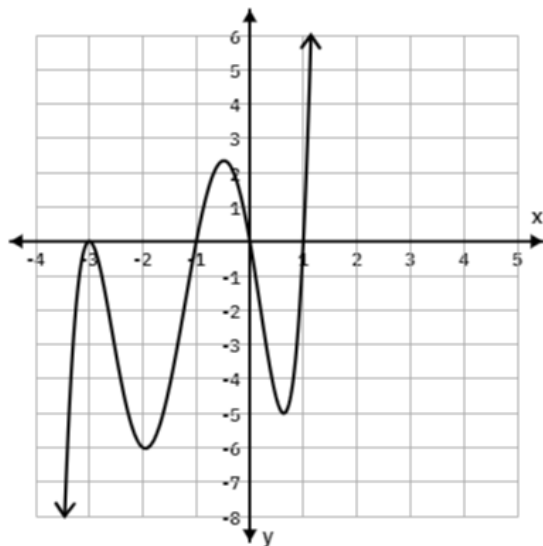


Figure 6: reference attached

The graph of $y = f(x)$ is shown. Which of the following best represents the function $f(x)$?

- (A) $f(x) = x(x - 1)(x + 1)(x + 3)^2$
- (B) $f(x) = x(x + 1)(x - 1)(x + 3)$
- (C) $f(x) = (x - 3)^2(x - 1)(x + 1)$
- (D) $f(x) = (x + 3)^2(x + 1)(x - 1)$

Answer:

□

41. **Algebraic Expression Simplification 2** (10 points)

Which of the following expressions is equivalent to $\frac{x+3}{x^2-4x+3} + \frac{2}{x-1}$?

- (A) $\frac{1}{x-1}$
- (B) $\frac{3}{x-3}$
- (C) $\frac{3x+3}{x^2-4x+3}$
- (D) $\frac{x+5}{x^2-3x+2}$

Answer:

□

42. **System of Equations: No Solution** (10 points)

$$3x + by = 36ax - 7y = 24$$

If the given system of equations has no solution, what is the value of ab ?

Answer:



43. **Linear Inequality Point Condition** (10 points)

The point (a, a) satisfies the linear inequality $y \leq mx + 1$, and $a > 0$. The value of m must be greater than or equal to what value?

(A) $\frac{1}{a} - 1$

(B) $a - 1$

(C) $1 - \frac{1}{a}$

(D) $1 - a$

Answer:



44. **Projectile Motion: Maximum Height** (10 points)

A ball is kicked off of a platform modeled by the equation $y = -5x^2 + bx + 10$, such that y is height in feet, x is seconds, and b is a constant. The ball reaches a maximum height of 11.25 feet, and it takes 2 seconds to hit the ground. How many seconds does it take for the ball to reach its maximum height?

Answer:

